ENGINEERS WALLAH

CHEMISTRY

Lecture XIII

Periodic Classification of Elements

'BRIEF'



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TOPICS *to be covered*

1. Dobereiner Triads and Drawbacks
2. Newlands' Laws of Octaves and Drawbacks
3. Mendeleev's Periodic Table and Drawbacks
4. Modern Periodic Table and Trends

Knowledge Ride On



Dobereiner Triads and Drawbacks

Knowledge Ride On



Newlands' Laws of Octaves and Drawbacks

Knowledge Ride On



Mendeleev's Periodic Table and Achievements

Knowledge Ride On



Modern Periodic Table and Trends

Knowledge Ride On



Efficiency Hacks

Glossary

Periodic table (आवर्त सारणी)
Atomic radius (परमाणु त्रिज्या)
Metallic character (धात्विक चरित्र)
Non-metallic character (गैर-धात्विक चरित्र)
Electronegativity (वैद्युतीयऋणात्मकता)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

The chemical formula of acetone is

SSC JE 2018

A $(CH_3)_2O$

B $(H_3)_2CO$

C CH_3CO

D $(CH_3)_2CO$

Chemical name

Propan-2-one or Propanone

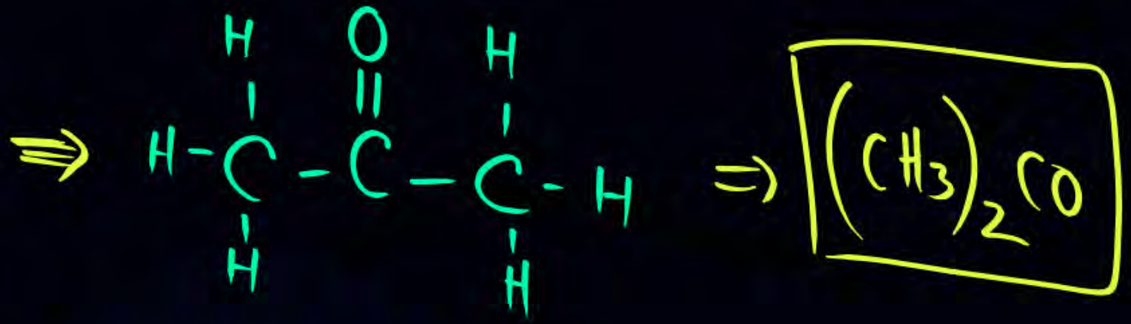
3 carbon atoms

Carbonyl

F.G. is ketone [Carbonyl]

single bond

Ketone → R-C(=O)-R



Question

Vinegar is an aqueous solution of:

→ water is solvent

SSC JE 2018

A. Formic acid

B. Acetic acid ✓

C. Lactic acid

D. Malic acid

Ethanoic acid / Acetic acid → CH_3COOH

↓

Diluted form [92-95% water]
[8% 5% acetic acid]

↓

VINEGAR (सिरका)

Question

How many hydrogen atoms are present in chloropropane?

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A 5

B 7

C 9

D 3

→ Haloalkane / Alkyl halide

→ General formula: $R-X$ | $C_nH_{2n+1}-X$

Halogens (F, Cl, Br, I)

C_3H_7-Cl

chlorine

n=3

German chemist

Concept Bites – Dobereiner's Triads and Drawbacks

Gyaan I

Classification of elements

↓

'GROUPS' [All Elements have same property]

↓

'To make their study easier'

Gyaan II

→ First classification of elements → Lavoisier

→ Two groups: METALS and NON-METALS

→ 30 elements known at that time.

Doberiener's Triads → 'Group of 3 elements'

'IDEA'

(i) Dobereiner arranged three elements in a group (triad) with *similar chemical properties* in order of *increasing atomic masses*.

↓

1817

↓

(ii) He observed a quantifiable relationship between those elements.

Li	→ 6.9
Na	→ 23
K	→ 39

(i) Same chemical properties (v)

(ii) (↑) atomic mass

$= \frac{6.9 + 39}{2} = 22.95 \simeq 23$

Elements	Chemical Symbol	Atomic Mass (in u)	Arithmetic Mean (in u)
Lithium	Li	6.9	22.95
Sodium	Na	23	
Potassium	K	39	
Calcium	Ca	40.1	88.7
Strontium	Sr	87.6	
Barium	Ba	137.3	
Chlorine	Cl	35.5	81.2
Bromine	Br	79.9	
Iodine	I	126.9	
Sulphur	S	32.06	79.83
Selenium	Se	78.96	
Tellurium	Te	127.60	
Iron	Fe	55.8	57.25
Cobalt	Co	58.9	
Nickel	Ni	58.7	

Triad I st

II nd

III rd

IV th

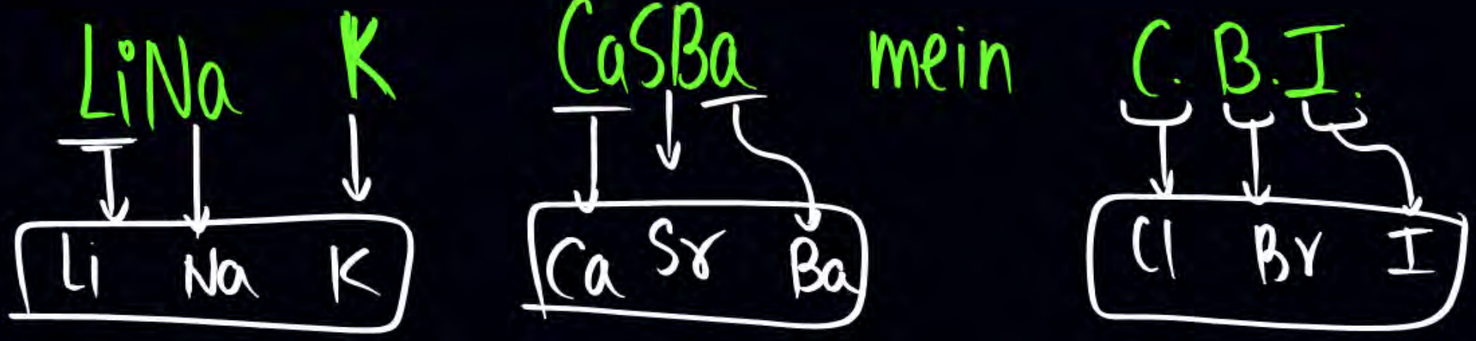
V th

1817

Rest other triads in 1829

Trick to Remember Dobereiner's Triad

TRICK Ist → LiNa K CaSBa mein C. B. I.



TRICK IInd → S.S.T. mein pdhte hai INC ke baare mein



Drawbacks of Dobereiner's Triad

'95' elements were known

(i) It failed to arrange all elements with similar chemical properties into triads.

N → 14
P → 31
As → 74.9

(I) – Similar chemical property (✓)
– (↑) atomic mass (✓)

(II) $\frac{14+74.9}{2} = 44.95 \neq 31$

(ii) Some elements that follow the quantifiable relationships do not have similar chemical properties.

C → 12
N → 14
O → 16

(I) – Similar chemical property (✗)
– (↑) atomic mass (✓)

(II) $\frac{12+16}{2} = \frac{28}{2} = 14$ = atomic mass of nitrogen

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

If two members of a Dobereiner's triad are calcium and strontium, the third element of this triad is _____.

A Magnesium

B Barium

C Caesium

D Sodium

Ca ✓
Sr ✓
Ba

Question

Which among the following is not a Dobereiner's triads?

A. Li, Na, K

B. Ca, Sr, Ba

C. Cl, Br, I

D. C, N, O

British chemist

Concept Bites – Newlands' Laws of Octaves and Drawbacks

Newlands's Laws of Octaves

(i) In Indian musical system there are seven notes, i.e. *sa, re, ga, ma, pa, dha, ni* while in the western musical system these are called *do, re, mi, fa, so, la, ti. do*

(ii) Newlands' found that every eighth element possesses properties similar to that of first one; when arranged in order of increasing atomic masses.

Newlands' Laws of Octaves

Condensed form

sa (do)	re (re)	ga (mi)	ma (fa)	pa (so)	da (la)	ni (ti)
H	Li	Be	B	C	N	O
F	Na	Mg	Al	Si	P	S
Cl	K	Ca	Cr	Ti	Mn	Fe
Co and Ni	Cu	Zn	Y	In	As	Se
Br	Rb	Sr	Ce and La	Zr	—	—

Drawbacks of Newlands' Laws of Octaves

(i) Newlands' law of octaves was applicable only to lighter elements having atomic masses upto 40 u, i.e. Calcium. After Calcium every eighth element does not show properties similar to that of first element.

'Overconfidence'

Last known element
↓
Thorium (Th)

(ii) Newlands' assumed that only 56 elements existed in nature and no new elements would be discovered in future.

Drawbacks of Newlands' Laws of Octaves

1860s'

(iii) Newlands' adjusted two elements in the same slot; put some unlike elements under the same column and like elements in different coloumns.

(I) Co-Ni

(II) Co-Ni are placed with H, F, Cl, Br

(III) Co-Ni were placed away from Fe.

(iv) When noble gases were discovered around the year 1900, the properties of every eight element were no longer similar to the first one. Actually, it was the first and ninth element that had similar properties.

1st	Li
2	Be
3	B
4	C
5	N
6	O
7	F
8	Ne
9th	Na

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which element shows similar chemical properties like boron according to Newlands' laws of octaves?

→ noble gases weren't discovered.

A. Lithium

B. Sodium

C. Aluminium ✓

D. Carbon

B	I
C	II
N	III
O	IV
F	V
Na	VI
Mg	VII
Al	VIII

Question

If 'Li' is the first element then which among the following element holds similar properties to it in accordance with Newlands' law of octaves?

A. F

B. Na

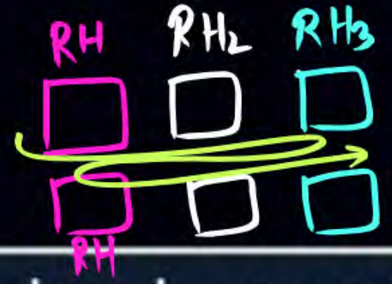
C. Ne

D. Mg

Li 1st
Be 2
B 3
C 4
N 5
O 6
F 7
Na 8th

Concept Bites – Mendeleev's Periodic Table and Achievements

Mendeleev's Periodic Table



One of the most important contributor to the early development of *periodic table* was *Dmriti Ivanovich Mendeleev*. → Russian chemist

- periodic table → chart or table that contains elements with similar chemical properties in vertical columns called groups.
- Dmriti Ivanovich Mendeleev → 'Father of periodic table'



'IDEA'

Mendeleev took 63 elements known at that time and wrote down the *symbol and properties (physical and chemical)* of each element on a card.

- physical and chemical → 'oxides & hydrides formula of elements'
  - 'oxides & hydrides formula of elements' → elements that have some formula
  - elements that have some formula → Placed in some group
  - Placed in some group → Elements automatically got arranged in order of (↑) atomic mass

Features of Mendeleev's Periodic Table

(I) Horizontal rows → 'Periods'

7 ✓

(II) Vertical columns - 'GROUPS' →

'8' ✓

Group	I	II	III	IV	V	VI	VII	VIII
Oxide Hydride	R_2O RH	RO RH_2	R_2O_3 RH_3	RO_2 RH_4	R_2O_5 RH_3	RO_3 RH_2	R_2O_7 RH	RO_4
Periods ↓	A B	A B	A B	A B	A B	A B	A B	Transition series
1	H 1.008							
2	Li 6.939	Be 9.012	B 10.81	C 12.011	N 14.007	O 15.999	F 18.998	
3	Na 22.99	Mg 24.31	Al 29.98	Si 28.09	P 30.974	S 32.06	Cl 35.453	
4 First series:	K 39.102	Ca 40.08	Sc 44.96	Ti 47.90	V 50.94	Cr 50.20	Mn 54.94	Fe 55.85 Co 58.93 Ni 58.71
Second series:	Cu 63.54	Zn 65.37	Ga 69.72	Ge 72.59	As 74.92	Se 78.96	Br 79.909	
5 First series:	Rb 85.47	Sr 87.62	Y 88.91	Zr 91.22	Nb 92.91	Mo 95.94	Tc 99	Ru 101.07 Rh 102.91 Pd 106.4
Second series:	Ag 107.87	Cd 112.40	In 114.82	Sn 118.69	Sb 121.75	Te 127.60	I 126.90	
6 First series:	Cs 132.90	Ba 137.34	La 138.91	Hf 178.49	Ta 180.95	W 183.85		Os 190.2 Ir 192.2 Pt 195.09
Second series:	Au 196.97	Hg 200.59	Tl 204.37	Pb 207.19	Bi 208.98			

7

Th U

Important to Remember

Mendeleev's Periodic Law

⇒ Physical and chemical properties of elements are the periodic function of their atomic masses.

OR Same physical & chemical property reoccur after regular intervals of atomic mass.

⇒ Prediction of new elements. ✓

eka- Bhi Al Si hai (In future)

eka- boron → Scandium (Sc)

eka-aluminium → Gallium (Ga)

eka-silicon → Germanium (Ge)

eka → Sanskrit numeral → 'first'

B 10.81	C 12.011
Al 29.98	Si 28.09
	Ti 47.90
Y 88.91	Zr 91.22
In 114.82	Sn 118.69

Practice and PYQ's Wallah



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Question

According to Mendeleev's ' Periodic Law, the elements were arranged in the periodic table in the order of:

A Increasing atomic number

B decreasing atomic number

C Increasing atomic mass

D decreasing atomic mass

Concept Bites – Modern Periodic Table and Trends

→ prepared by Bohr

→ Also, known as Bohr's table.

Modern Periodic Law

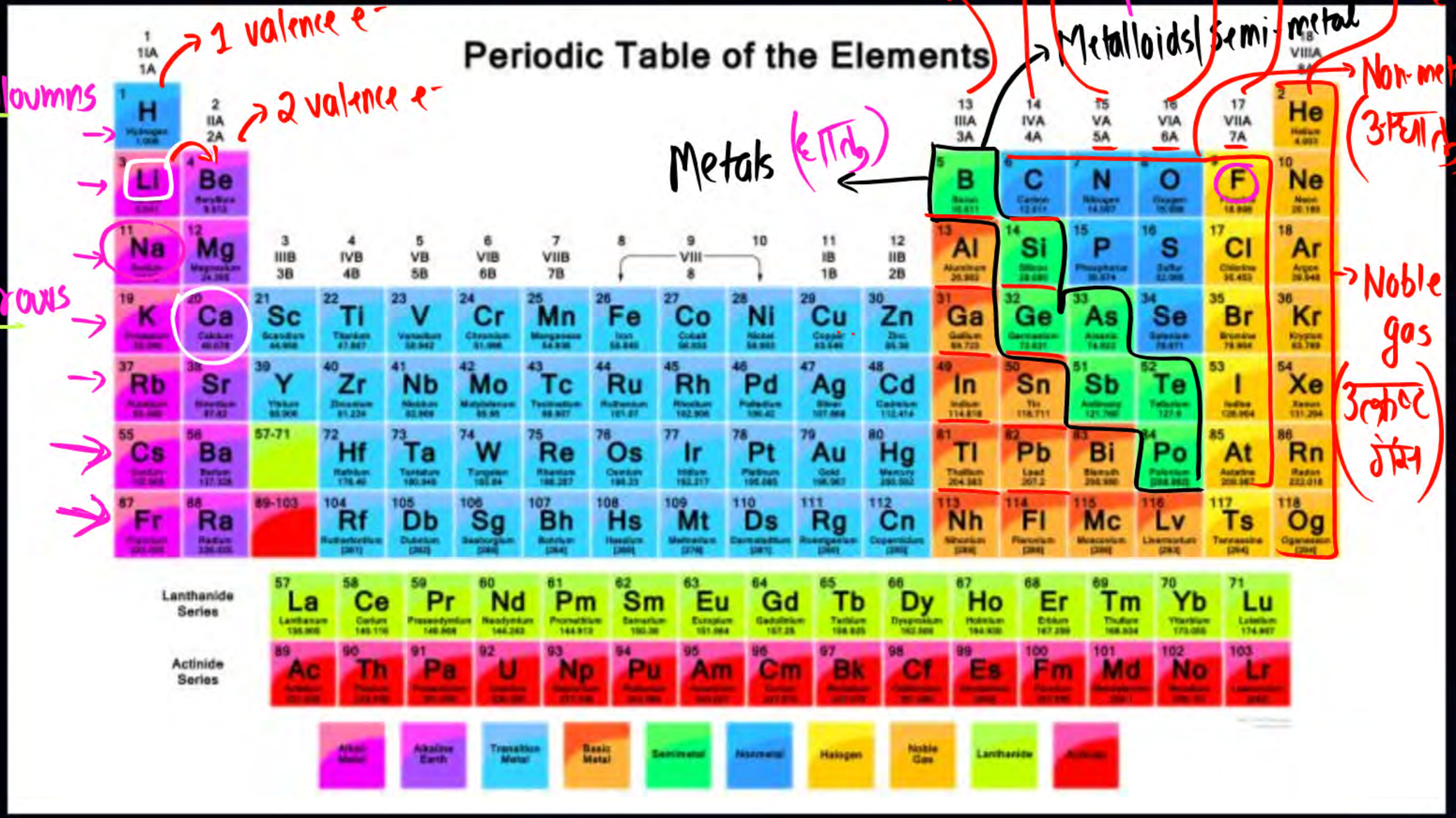
Physical and chemical properties of elements are the periodic function of their atomic numbers.

		4	5	6	7	8	9	10
3	Li	Be	B	C	N	O	F	Ne
11	Na	Mg	Al	Si	P	S	Cl	Ar
19	K							

Modern Periodic Table

18 vertical columns → 'GROUPS'

7 horizontal rows → 'PERIODS'



Name → water-soluble base

GROUP IA / 1 → Alkali metals ⟶ On run with water they form water-soluble bases.

GROUP IIA / 2 → Alkali earth metals → water-soluble oxides (alkali) obtained from earth

GROUP IIIA / 13 → Boron family / Icosagens [B-12 - Structure is icosahedron], triels [3 valence e⁻]

GROUP IVA / 14 → Carbon family / Crystallogens [Carbon- crystalline], tetrels [4 valence e⁻]

GROUP VA / 15 → Nitrogen family / fertiliser group [N, P] / pnictogens [suffocating compounds], pentels → 5 valence e⁻

GROUP VIA / 16 → Oxygen family OR chalcogens [ore- forming]

-"- VIIA / 17 → Halogens [salt producer]

-"- VIIIA / 18 → Noble gas / Rare gas / Inert gas / Zero group

Position of Elements in Modern Periodic Table

	Period no.	Group no.
K L M Na 2 8 1 A. No. (11)	last e^- in 3rd shell	1 valence e^- ⇒ Group 1
K L M N Ca 2 8 8 2 A. No. → 20	last e^- in 4th shell	2 valence e^- → Group II

Period no.

Group no.

A. No. → 9

F

K	L
2	7

Last e⁻ in 2nd shell

7 valence e⁻ → Group 17

Practice and PYQ's Wallah



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Question

Which of the following is incorrect about modern periodic table?

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A. Rows of modern periodic table are called periods.

B. Columns of modern periodic table are called groups.

C. From left to right across a period, each element has one more proton than the element before it.

D. Modern periodic table has 15 columns.

Question

Identify the element that does not belong to period 3 of modern periodic table.

SSC JE 2022

A Hydrogen ✓ → Period 1

B Aluminium

C Magnesium

D Sodium

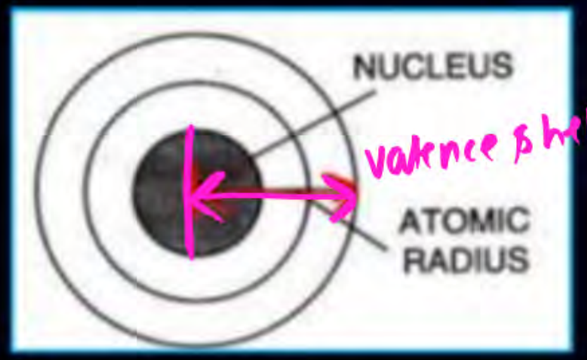
Period no.

H | K 1 | Last e⁻ in 1st shell

	K	L	M	
Na	2	8	1	→ Last e⁻ in 3rd shell
Mg	2	8	2	
Al	2	8	3	

3rd PERIOD

Trends in Atomic Radius – Across a Period [Left to Right]

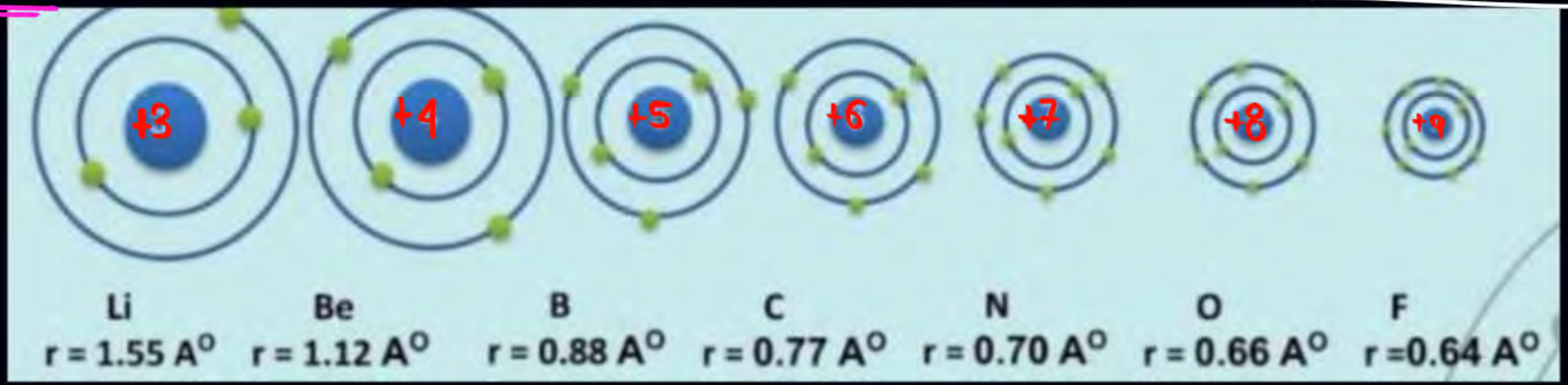


(I) Number of shells → Same

(II) Nuclear charge → (↑) in steps of 1

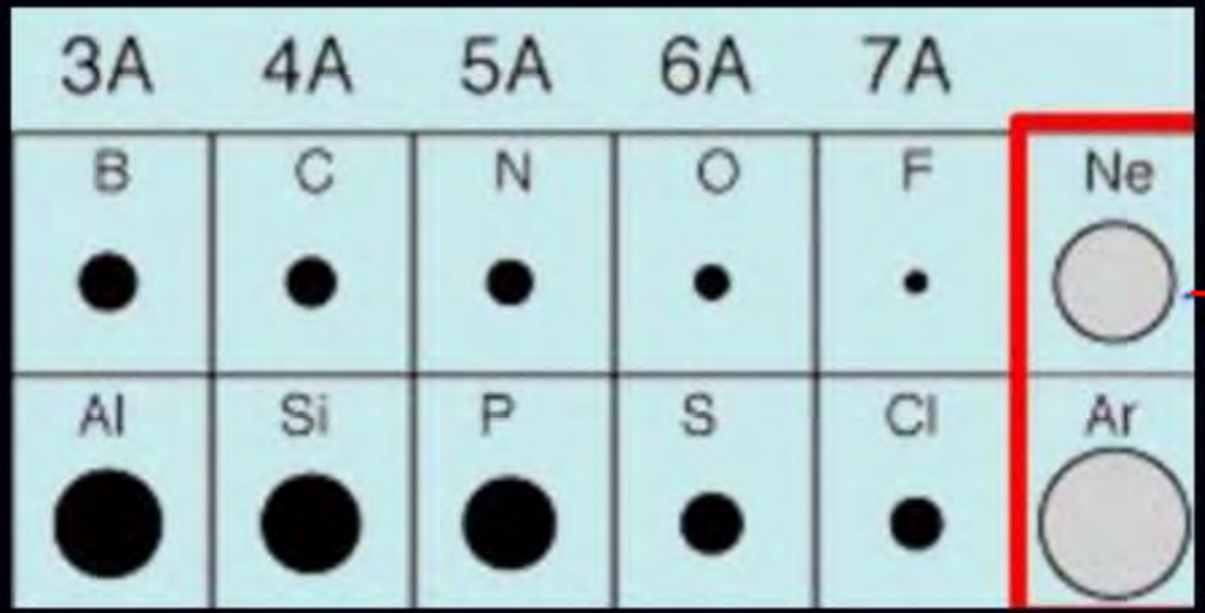
Atomic radius (↓)

Period 2nd



Trends in Atomic Radius – *Across a Period*

Atomic size of noble gas does not follow the trend across a period.



→ largest atomic radius in 2nd period

fully filled valence shell → electronic repulsion (↑)

Trends in Atomic Radius – Down the Group (Top to Bottom)

Atomic radius (↑)

Number of shells

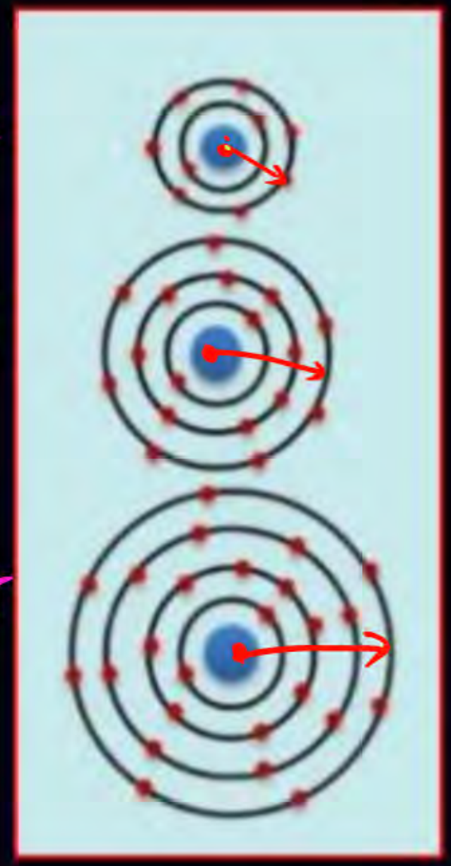
(↑) in steps of 1

distance btw nucleus & outermost shell (↑)

Nuclear charge

'also increases'

effect of this is overpowered by



Trends in Metallic Character – Across a Period (Left to Right)

'easily donate electrons'

- no. of Shells → remains same
- nuclear charge (↑) → electrons are attracted by nucleus & are not easily donate → Metallic character (↓) & Non-metallic character (↑)

Metals | Non-metals



Trends in Metallic Character -- Down the Group (Top to Bottom)

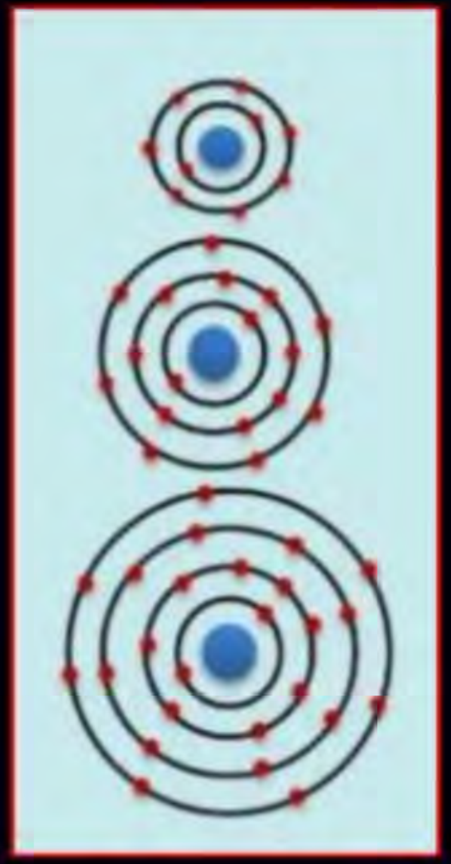
Increases down the group

→ Non-metallic character → (↓)

Most non-metallic element → Fluorine (F)

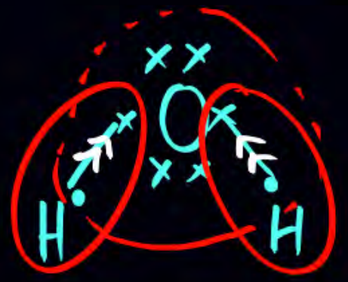
Most metallic element (naturally) → Caesium (Cs)

Most metallic element [Radioactive] → Francium (Fr)



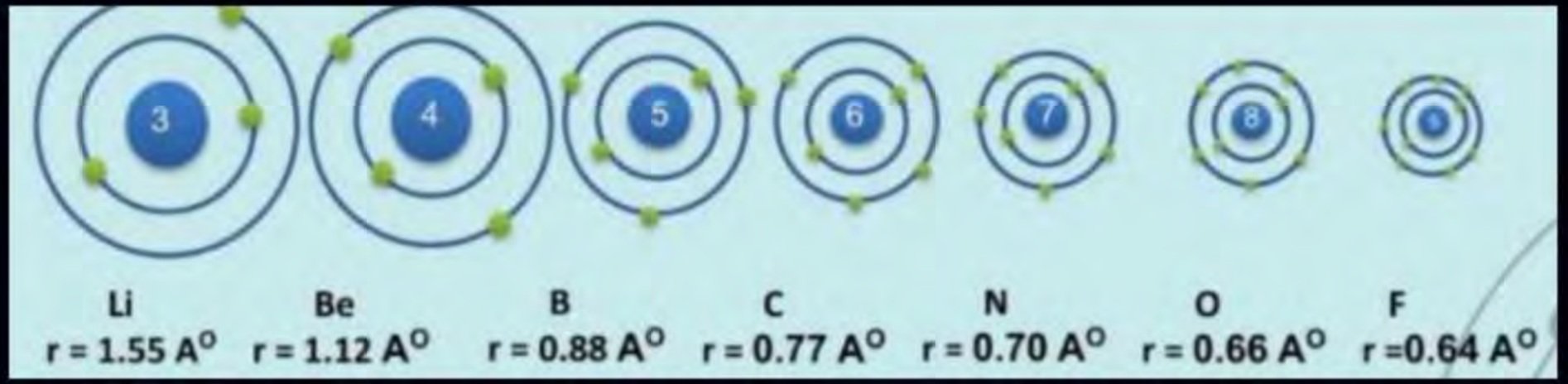
Trends in Electronegativity – Across a Period [Left to Right]

tendency to attract shared pair of electron towards itself



Electronegativity → Left to right → 'INCREASE'

Electropositivity → — " — → 'DECREASE'



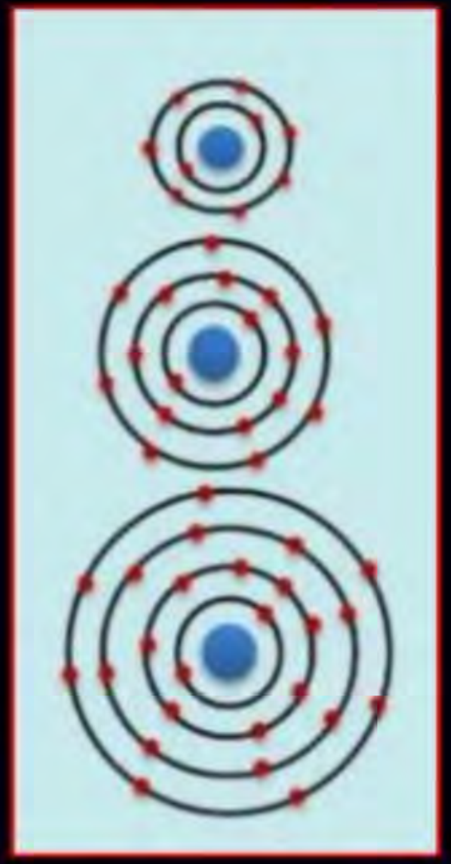
Trends in Electronegativity – Down the Group (Top to bottom)

ELECTRONEGATIVITY → Decrease

ELECTROPOSITIVITY → INCREASE

Most electronegative element → Fluorine (F)

-"- electropositive -"- ⟶ Caesium (Cs) or Francium (Fr) (if radioactive element is considered)



Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following is most electronegative element among Arsenic, Nitrogen and Phosphorus?

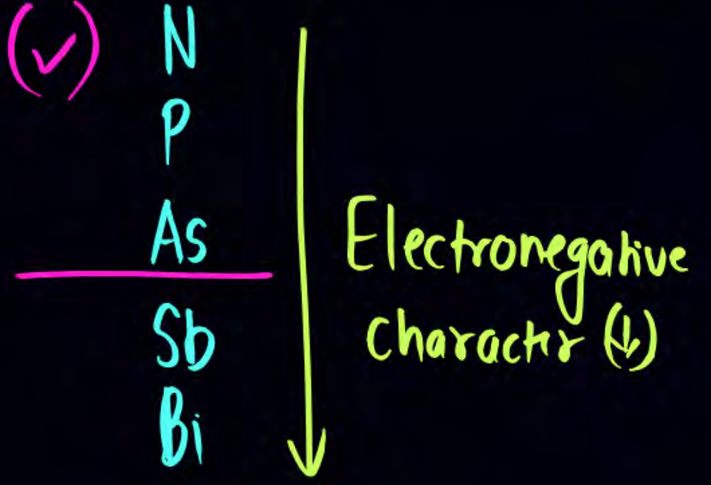
SSC JE 2017

A. Nitrogen

B. Phosphorus

C. Arsenic

D. All these elements have equal electronegativity



Question

Which of the following group 14 element has highest electronegativity?

SSC JE 2016

A Silicon

B Carbon

C Germanium

D Tin

Carbon (C)
Si
Ge
Sn
Pb

Electronegative character (↓)

Efficiency Hacks!

Eisenhower Matrix to Increase Productivity

Eisenhower Matrix is a simple decision-making tool that helps you make the distinction between tasks that are important, not important, urgent, and not urgent.




#sbsathhai
#pwsathhai
Thank
You !